

HOT DOG POINT-TO-SALE SYSTEM

Technical Presentation

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KATA OVERVIEW

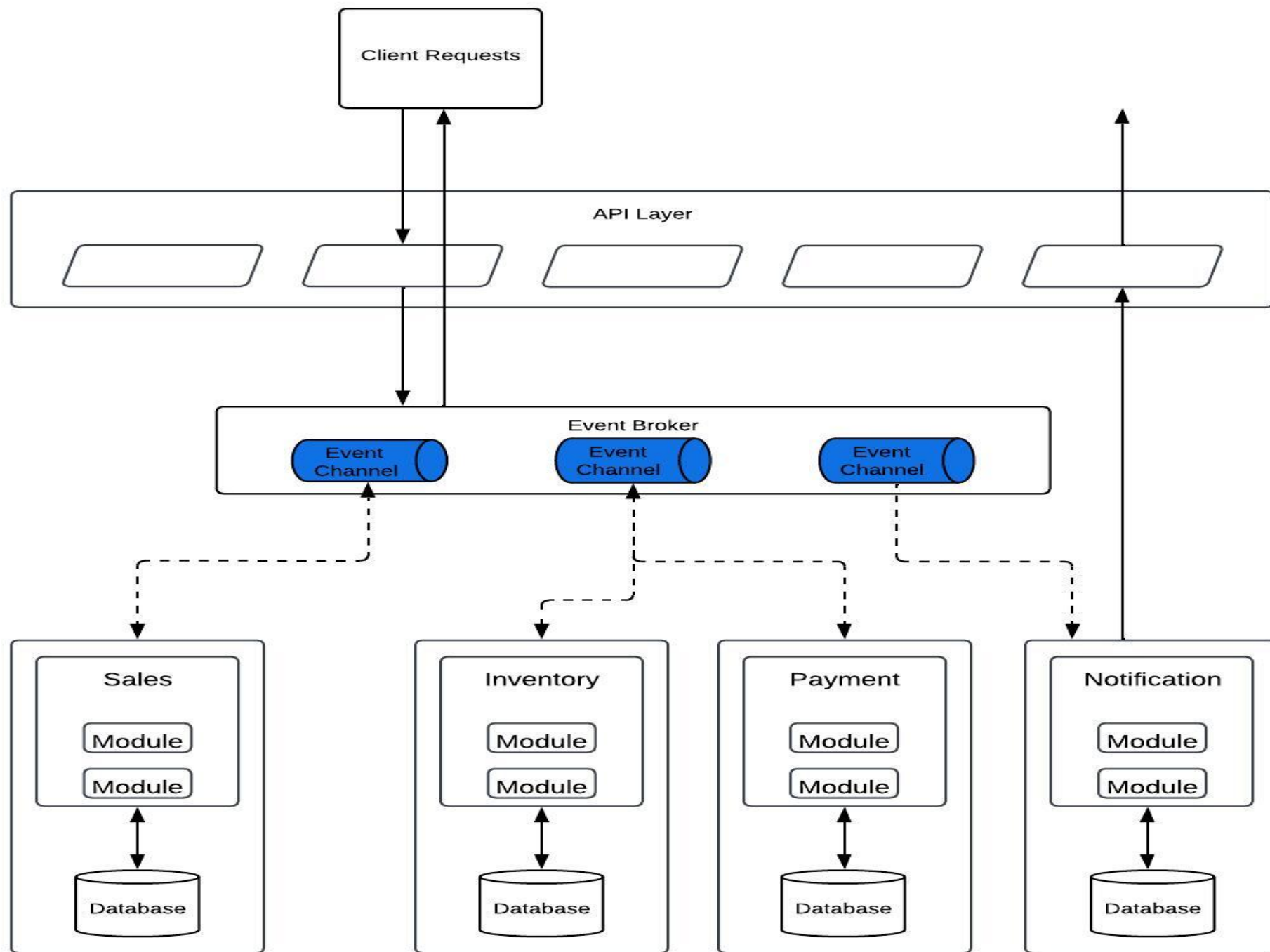
- A local hot dog stand merchant needs a lightweight, scalable, and efficient point-to-sale(POS) system for fifty stand operators serving thousands of customers. The system needs to have sales tracking, inventory updates, social-media integration, and discount application functionalities, and it should be portable and easy to use on the street. This system also needs to support data exportation for accounting purpose.

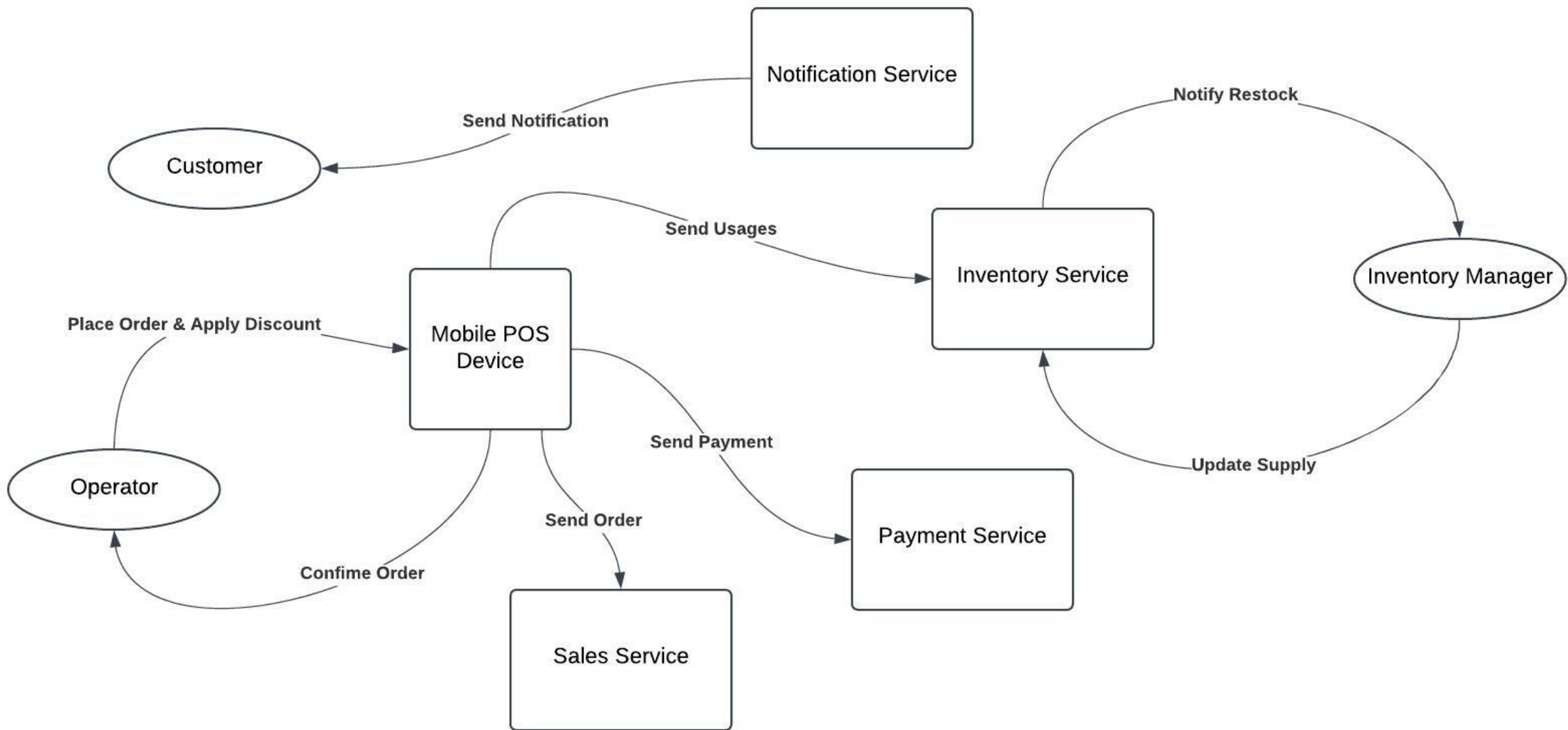
ARCHITECTURALLY INTERESTING BITS

- Consider a long-term solution and **scalability** for future business grows.
- The time of completion is important.
- Don't need to worry about budget.
- Design to build to reduce manual effort significantly.
- Lightweight design because the system needs to run on compact devices like tables or mobile phones.
- Real-time inventory updates require data synchronization.
- Social Media Integration.

PROPOSED SOLUTION – A COMBINATION OF
MICROSERVICES AND EVENT-DRIVEN ARCHITECTURE

- A combination of microservices and event-driven architecture ensure scalability, real time updates, and elasticity.





OTHER SOLUTION CONSIDERED BUT DISCARD

- Layered Architecture
- Microservices Architecture Alone

DEPLOYMENT ENVIRONMENT

- Client Application: On tables or mobile phones.
- Cloud Infrastructure: Leverage third-party cloud service for hosting microservices.
- Social Media Integration: API calls to popular social media platforms.

FITNESS FUNCTIONS

- Performance Testing
- Latency Testing
- Scalability Testing
- Social Media Functionality Testing

FAILURE CASES AND MITIGATION

- Different table or mobile systems failure: IOS vs. Android.
- Inventory mismatch: Implement transaction logging and conflict resolution.
- Internet outages: Prevent internet outages data lost with offline-first pattern.
- Data loss in the cloud: Regular backups and redundancy for databases.
- API rate limits for social media: Use batching strategies and monitor API usage limits.

STATES

- Normal state: All services operational, transactions processed within SLAs.
- Warning state: Delays in inventory updates and notification triggers.
- Alarm state: Not functional and service outages.

EXPECTED DATA GROWTH AND IMPACTS

- Expect the number of operators increase, which means transaction increase.
- Increased data storages and API usages, which requires the system to be able to scale and handle high volume traffic.

BUSINESS CONTINUITY NEEDS

- High Availability
- Disaster Recovery
- Support and Maintenance